

Z8440/1/2/4, Z84C40/1/2/3/4

SERIAL INPUT/OUTPUT CONTROLLER

FEATURES

- Two independent full-duplex channels, with separate control and status lines for modems or other devices.
- Data rate in the x1 clock mode of 0 to 2.0M bits/second with a 10 MHz clock.
- NMOS version for cost sensitive performance solutions, CMOS version for the designs requiring low power consumption.
- NMOS Z0844x04 – 4 MHz, Z0844x06 – 6.17 MHz (Where x is the designator for the bonding option; 0, 1, 2 or 4)
- CMOS Z84C4x06 – DC to 6/7 MHz, Z84C4x08 – DC to 8 MHz, Z84C4x10 – DC to 10 MHz (Where x is the designator for the bonding option, 0, 1, 2, or 4)
- 6 MHz version supports 6.144 MHz CPU clock operation.
- Asynchronous protocols: everything necessary for complete messages in 5, 6, 7, or 8 bits/character. Includes variable stop bits and several clock-rate multipliers; break generation and detection; parity; overrun and framing error detection.
- Synchronous protocols: everything necessary for complete bit- or byte-oriented messages in 5, 6, 7, or 8 bits/character, including IBM Bisync, SDLC, HDLC, CCITT-X.25 and others. Automatic CRC generation/checking, sync character and zero insertion/deletion, abort generation/detection, and flag insertion.
- Receiver data registers quadruply buffered, transmitter registers doubly buffered.
- Highly sophisticated and flexible daisy-chain interrupt vectoring for interrupts without external logic.

GENERAL DESCRIPTION

The Z80 SIO (hereinafter referred to as the Z80 SIO or, SIO) Serial Input/Output Controller is a dual-channel data communication interface with extraordinary versatility and capability. Its basic functions as a serial-to-parallel, parallel-to-serial converter/controller can be programmed by a CPU for a broad range of serial communication applications.

The device supports all common asynchronous and synchronous protocols, byte- or bit-oriented, and performs all of the functions traditionally done by UARTs, USARTs, and synchronous communication controllers combined, plus additional functions traditionally performed by the CPU. Moreover, it does this on two fully-independent channels, with an exceptionally sophisticated interrupt structure that

allows very fast transfers.

Full interfacing is provided for CPU or DMA control. In addition to data communication, the circuit can handle virtually all types of serial I/O with fast, or slow, peripheral devices. While designed primarily as a member of the Z80 family, its versatility makes it well suited to many other CPUs.

The Z80 SIO uses a single +5V power supply and the standard Z80 family single-phase clock. The SIO/0, SIO/1, and SIO/2 are packaged in a 40-pin DIP, the SIO/4 is packaged in a 44-pin PCC and the SIO/3 is packaged in a 44-pin QFP. Note that SIO/3 is only available in CMOS and in QFP package.

PIN DESCRIPTION

Figures 1 through 6 illustrate the three 40-pin configurations (bonding options) available in the Z80 SIO (hereafter referred to as SIO or Z80 SIO). The constraints of a 40-pin package make it impossible to bring out the Receive Clock ($\overline{\text{RxC}}$), Transmit Clock ($\overline{\text{TxC}}$), Data Terminal Ready ($\overline{\text{DTR}}$) and Sync ($\overline{\text{SYNC}}$) signals for both channels. Therefore, either Channel B lacks a signal or two signals are bonded together.

- Z80 SIO/2 lacks $\overline{\text{SYNCB}}$
- Z80 SIO/1 lacks $\overline{\text{DTRB}}$

- Z80 SIO/0 has all four signals, but $\overline{\text{TxCB}}$ and $\overline{\text{RxCB}}$ are bonded together

The 44-pin package, the Z80 SIO/4 for PLCC package, and Z80 SIO/3 for QFP, has all options (Figure 7a and 7b).

The first bonding option above (SIO/2) is the preferred version for most applications. The pin descriptions are as follows:

B/ $\overline{\text{A}}$. Channel A or B Select (input, High selects Channel B). This input defines which channel is accessed during a data

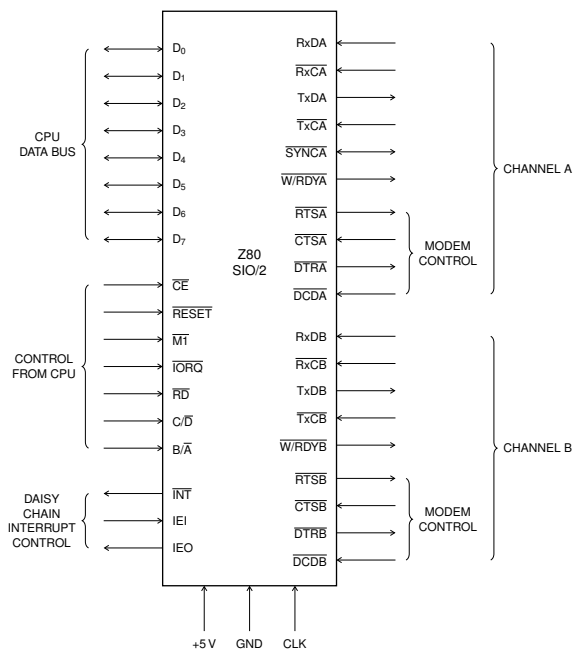


Figure 1. Pin Functions

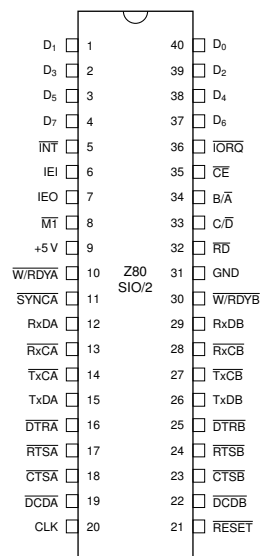


Figure 2. 40-pin Dual-In-Line Package (DIP), Pin Assignments